

II-Probabilistic Approaches Likelihood, empirical distributions and learning

Foundations of Neuro-Symbolic AI

Alex Goessmann

University of Applied Science Würzburg-Schweinfurt (THWS)

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Outline

- ▶ Log-likelihood
- ▶ Empirical Distributions
- ▶ Learning Bayesian networks

How to select a model given data?

Let us consider the typical **learning problem**:

- ▶ The distribution $\mathbb{P} [X_{[d]}]$ of variables $X_{[d]}$ is unknown
- ▶ We observe m samples

$$x_{[d]}^j = (x_0^j, \dots, x_{d-1}^j)$$

enumerated by $j \in [m]$.

We assume that each sample $x_{[d]}^j$ is drawn

- ▶ **i**ndependently from each other and
- ▶ **i**dentically **d**istributed by a distribution $\mathbb{P} [X_{[d]}]$.

This assumption is abbreviated by **i.i.d.**.

The probability of drawing a dataset $X_{[d]}^{[m]}$ from a distribution $\mathbb{P} [X_{[d]}]$ is then

$$\prod_{j \in [m]} \mathbb{P} [X_{[d]} = x_{[d]}^j] .$$

This is the **likelihood** of $\mathbb{P} [X_{[d]}]$.

The likelihood principle

We make the hypothesis that the distribution is from a set Γ of possible distributions. A natural choice would be to choose the distribution maximizing the likelihood

$$\operatorname{argmax}_{\mathbb{P}[X_{[d]}] \in \Gamma} \left(\prod_{j \in [m]} \mathbb{P} \left[X_{[d]} = x_{[d]}^j \right] \right) .$$

Sometimes we **regularize**, that is prefer simple distributions over more complex ones.

Likelihood principle

The selection of a distribution should depend on the data $X_{[d]}^{[m]}$ only through the likelihood

$$\prod_{j \in [m]} \mathbb{P} \left[X_{[d]} = x_{[d]}^j \right] .$$

Negative log-likelihood

We simplify the likelihood by taking the log and averaging. This is the **log-likelihood**

$$\frac{1}{m} \ln \left[\prod_{j \in [m]} \mathbb{P} \left[x_{[d]}^j \right] \right] = \frac{1}{m} \sum_{j \in [m]} \ln \left[\mathbb{P} \left[x_{[d]}^j \right] \right] .$$

The maximum likelihood method corresponds then with the minimization of the **negative log-likelihood (NLL)**

$$\operatorname{argmin}_{\mathbb{P}[X_{[d]}] \in \Gamma} \frac{1}{m} \sum_{j \in [m]} - \ln \left[\mathbb{P} \left[X_{[d]} = x_{[d]}^j \right] \right] .$$

Example: Thumbtack

Let there be a variable X describing whether a thumbtack lands on the head ($x = 1$) or the tail side ($x = 0$).

- We want to estimate the probability $p \in [0, 1]$ of head, that is the distribution:

$$\mathbb{P}[X] = \boxed{\begin{pmatrix} 1-p \\ p \end{pmatrix}}$$

$\downarrow x$



- We throw the thumbtack m times and observe m_h heads. The negative log likelihood score is then

$$-\ln[1-p] \cdot \frac{(m - m_h)}{m} - \ln[p] \cdot \frac{m_h}{m}.$$

We find the minimum by differentiating

$$0 = \frac{\partial}{\partial p} \left[-\ln[1-p] \cdot \frac{(m - m_h)}{m} - \ln[p] \cdot \frac{m_h}{m} \right] = \frac{m - m_h}{m \cdot (1-p)} - \frac{m_h}{m \cdot p}.$$

Thus, the optimum is $p = \frac{m_h}{m}$.

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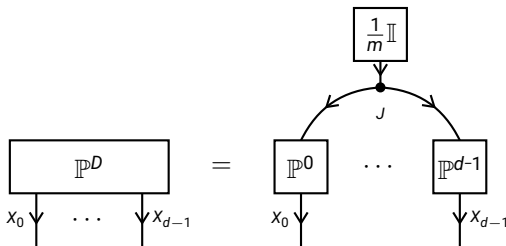
Empirical distributions

Let us understand the uniform choice of a sample from the dataset $x_{[d]}^{[m]}$ as a distribution

$$\mathbb{P}^D [X_{[d]}] = \frac{1}{m} \sum_{j \in [m]} \epsilon_{x_{[d]}^j} [X_{[d]}]$$

which we call **empirical distribution**.

We have the CP decomposition of the empirical distribution



where

$$\mathbb{P}^k [X_k | J] = \sum_{j \in [m]} \epsilon_{x_k^j} [X_k] \otimes \epsilon_j [J] .$$

Perspective of basis calculus

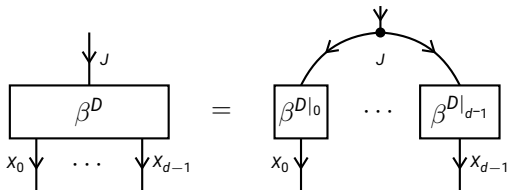
We understand the empirical distribution to samples $x_{[d]}^{[m]}$ as the basis encoding of the sample selecting map

$$D : [m] \rightarrow \bigtimes_{k \in [d]} [m_k]$$

defined for $j \in [m]$ by $D(j) = x_{[d]}^j$. To be more precise we have

$$\mathbb{P}^D [X_{[d]}] = \left\langle \beta^D [X_{[d]}, J], \frac{1}{m} \mathbb{I} [J] \right\rangle_{[X_{[d]}]}.$$

The CP decomposition of the empirical distribution is then based on the decomposition of the basis encoding into the basis encodings of its coordinate restrictions:



Cross-entropy

The **cross-entropy** between two distributions $\tilde{\mathbb{P}}[X_{[d]}]$ and $\mathbb{P}[X_{[d]}]$ is

$$\mathbb{H}[\tilde{\mathbb{P}}, \mathbb{P}] = \left\langle \tilde{\mathbb{P}}[X_{[d]}], -\ln[\mathbb{P}[X_{[d]}]] \right\rangle_{[\emptyset]}.$$

The negative log-likelihood is the cross-entropy between the empirical distribution $\mathbb{P}^D[X_{[d]}]$ and the candidate $\mathbb{P}[X_{[d]}]$:

$$\frac{1}{m} \sum_{j \in [m]} -\ln[\mathbb{P}[X_{[d]} = x_{[d]}^j]] = \left\langle \mathbb{P}^D[X_{[d]}], -\ln[\mathbb{P}[X_{[d]}]] \right\rangle_{[\emptyset]}$$

This is the contraction:

$$\mathbb{H}[\mathbb{P}^D, \mathbb{P}] = \begin{array}{ccc} & \boxed{-\ln[\mathbb{P}]} & \\ & | & \\ x_0 & \boxed{\dots} & x_{d-1} \\ & | & \\ & \boxed{\mathbb{P}^D} & \end{array} \leftarrow \text{Exploit tensor network decompositions!}$$

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- ▶ **Learning Bayesian networks**

Two learning tasks

Learning graphical models (Bayesian networks or Markov networks) typically consists in two tasks:

- ▶ **Structure learning:** Choose a hypergraph $\mathcal{G}$ for the graphical model.
- ▶ **Parameter estimation:** Find the optimal decorations by conditional probability distributions.

Outlook: Structure learning

A hypergraph $\mathcal{G}$ encodes some conditional independence statements about the distribution:

- ▶ Bayesian networks: Definition based on the parent sets
- ▶ Markov networks: Definition based on separation of variable clusters

Learning by independence tests

Learning $\mathcal{G}$ means deciding which conditional independence statements hold and choosing a suitable graph capturing most of them.

The task is hard:

- ▶ **Noise in the data:** Independencies in the generating distribution $\mathbb{P} [X_{[d]}]$ are not inherited by the empirical distribution $\mathbb{P}^D [X_{[d]}]$
→ Use robust hypothesis tests managing the noise
- ▶ **Superexponentially growing number of possible graphs:** There are $2^{d \cdot (d-1)}$ hypergraphs to d nodes
→ Need to apply further sparsity principles: Structure cores

Decomposing the negative log-likelihood

For a Bayesian network we have for any index tuple $x_{[d]}$

$$\ln [\mathbb{P} [X_{[d]} = x_{[d]}]] = \sum_{k \in [d]} \ln [\mathbb{P} [X_k = x_k | X_{\text{Pa}(k)} = x_{\text{Pa}(k)}]] .$$

The NLL thus decomposes into local terms

$$\begin{aligned} & \left\langle \mathbb{P}^D [X_{[d]}] , -\ln [\mathbb{P} [X_{[d]}]] \right\rangle_{[\emptyset]} \\ &= \left\langle \mathbb{P}^D [X_{[d]}] , \sum_{k \in [d]} -\ln [\mathbb{P} [X_k | X_{\text{Pa}(k)}]] \right\rangle_{[\emptyset]} \\ &= \sum_{k \in [d]} \left(\left\langle \mathbb{P}^D [X_{[d]}] , -\ln [\mathbb{P} [X_k = x_k | X_{\text{Pa}(k)} = x_{\text{Pa}(k)}]] \right\rangle_{[\emptyset]} \right) . \end{aligned}$$

The minimum can found separately for each k by the minimization problem

$$\operatorname{argmin}_{\mathbb{P} [X_k | X_{\text{Pa}(k)}]} \frac{1}{m} \sum_{j \in [m]} -\ln [\mathbb{P} [X_k = x_k | X_{\text{Pa}(k)} = x_{\text{Pa}(k)}]] .$$

Optimum by normalization of the empirical distribution

The optima for each conditional distribution is taken at

$$\begin{aligned} \operatorname{argmin}_{\mathbb{P}[X_k | X_{\text{Pa}(k)}]} & \left\langle \mathbb{P}^D, -\ln [\mathbb{P}[X_k = x_k | X_{\text{Pa}(k)} = x_{\text{Pa}(k)}]] \right\rangle_{[\emptyset]} \\ &= \left\langle \mathbb{P}^D [X_{[d]}] \right\rangle_{[X_k | X_{\text{Pa}(k)}]} \end{aligned}$$

This can be shown by the first order criterion (vanishing gradients).

Optimal parameters

Given a directed hypergraph, the local likelihood maximizers are the conditional probability distributions of the empirical distribution

$$\left\langle \mathbb{P}^D [X_{[d]}] \right\rangle_{[X_k | X_{\text{Pa}(k)}]} .$$